PE中节表IMAGE_SECTION_HEADER详（四）

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IMAGE_SECTION_HEADER

c AI写代码 | 复制 运行

```c
typedef struct _IMAGE_SECTION_HEADER {
+   BYTE Name[IMAGE_ SIZEOF_ SHORT_ NAME]; //ASCI字符串 可自定义只截取8个
    union {//没有对齐前的真实尺寸，即节的真实大小
    DWORD PhysicalAddress;
    DWORD VirtualSize;
    } Misc;
+   DWORD VirtualAddress;//在内存中的偏移地址,加上ImageBase才是在内存中的真正地址
+   DWORD SizeOfRawData;//节在文件中对齐后的尺寸
+   DWORD PointerToRawData;//节区在文件中的偏移
    DWORD PointerToRelocations,//调试相关
    DWORD PointerTol inenumbers;
    WORD NumberOfRelocations;
    WORD NumberOflinenumbers;
+   DWORD Characteristics;//节的属性
} IMAGE_ SECTION_ _HEADER, *PIMAGE_ _SECTION_ HEADER;
```

收起 ^

BYTE Name

各个节的名字，可更改

```
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00 ; .text...襦......
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00 ; .F..............
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00 ; .... ..`.rdata..
00000200h: 0E 20 00 00 00 60 00 00 00 22 00 00 00 4A 00 00 ; . ...`..."...J..
00000210h: 00 00 00 00 00 00 00 00 00 00 00 00 40 00 00 40 ; ............@..@
00000220h: 2E 64 61 74 61 00 00 00 E0 18 00 00 00 90 00 00 ; .data...?...?.
00000230h: 00 0C 00 00 00 6C 00 00 00 00 00 00 00 00 00 00 ; .....l..........
00000240h: 00 00 00 00 40 00 00 C0 2E 72 73 72 63 00 00 00 ; ....@..?rsrc...
00000250h: B4 01 00 00 00 B0 00 00 00 02 00 00 00 78 00 00 ; ?...?......x..
```

各个节偏移

- Misc ； 节的真实大小

```
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00 ; .text...襦......
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00 ; .F..............
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00 ; .... ..`.rdata..
```

- VirtualAddress; 在内存中的偏移地址,加上ImageBase才是在内存中的真正地址
  此文件在内存中偏移地址为1000

```
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00
```

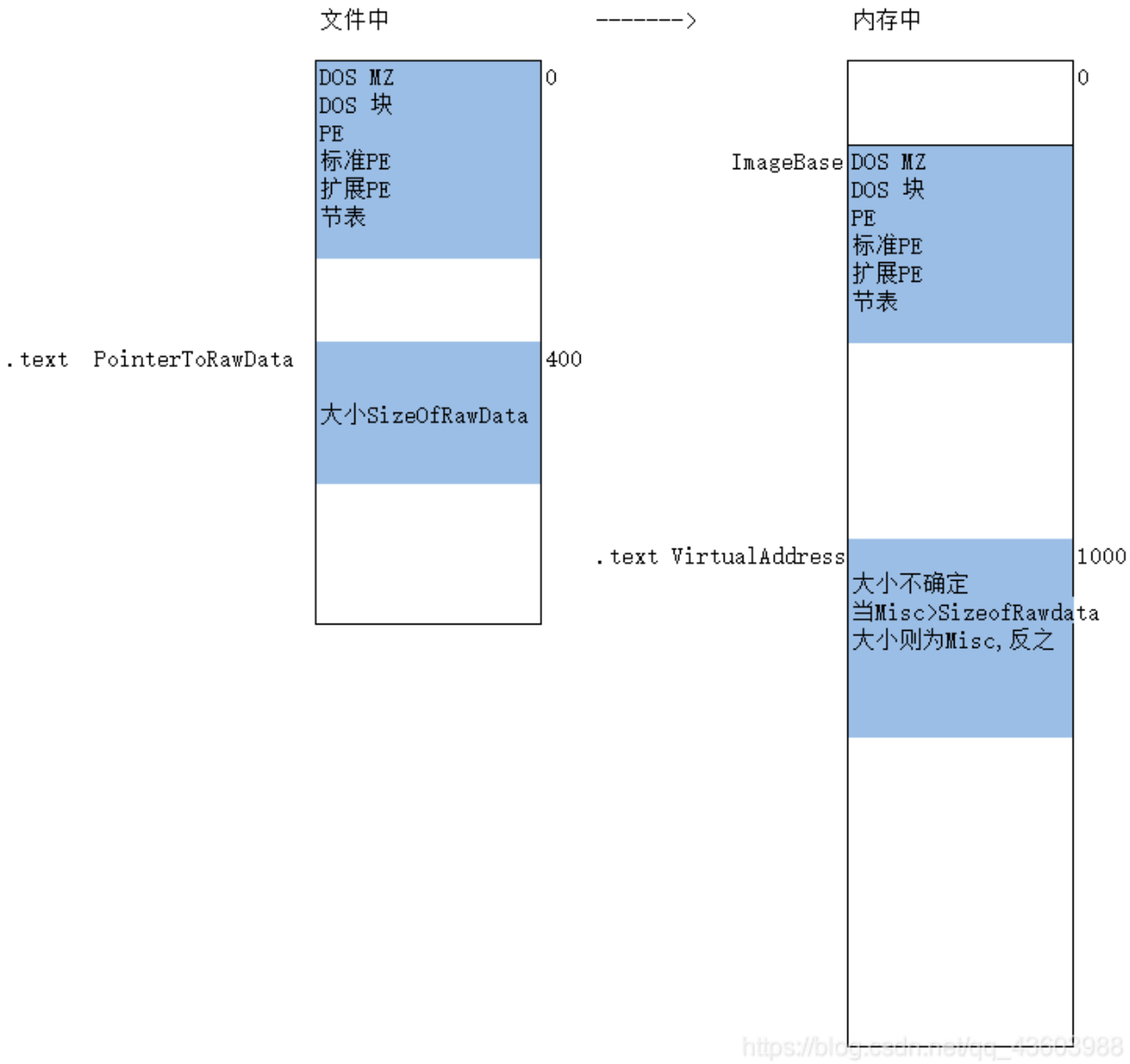
- SizeOfRawData;节在文件中对齐后的尺寸

```
000001c0h: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ; ................
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00 ; .text...襦......
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00 ; .F..............
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00 ; .... ..`.rdata..
```

- PointerToRawData;节区在文件中的偏移
  此文件偏移为400

```
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00 ; .text...襦......
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00 ; .F..............
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00 ; .... ..`.rdata..
```

由上可得他之间得对应关系入下图



DWORD Characteristics

```
000001d0h: 2E 74 65 78 74 00 00 00 D2 44 00 00 00 10 00 00 ; .text...襓......
000001e0h: 00 46 00 00 00 04 00 00 00 00 00 00 00 00 00 00 ; .F..............
000001f0h: 00 00 00 00 20 00 00 60 2E 72 64 61 74 61 00 00 ; .... ..`.rdata..
```

如图中20000060 因是小端存储

则转换为二进制：0110 0000 0000 0000 0000 0000 0010 0000

则对应的数据位如下表

数据位	常量符号	位为1时的含义
5	IMAGE SCN_ CNT_ CODE或0000020h	节中包含代码
6	IMAGE_ SCN_ CNT_ INITIALIZED_ DATA或0000040h	节中包含已初始化数据
7	IMAGE_ SCN_ CNT_ UNINITIALIZED_ DATA或00000080h	节中包含未初始化数据
8	IMAGE_ SCN_ LNK. OTHER或00000100h	保留供将来使用
25	IMAGE_ SCN_ MEM _DISCARDABLE或0200000h	节中的数据在进程开始以后将被丢弃，如.reloe
26	IMAGE_ SCN_ MEM_ NOT CACHED或04000000h	节中的数据不会经过缓存
27	IMAGE_ SCN_ MEM_ NOT_ PAGED或08000000h	节中的数据不会被交换到磁盘
28	IMAGE_ SCN_ MEM SHARED或10000000h	表示节中的数据将被不同的进程所共享
29	IMAGE_ SCN MEM_ EXECUTE或20000000h	映射到内存后的页面包含可执行属性
30	IMAGE_ SCN_ MEM READ或40000000h	映射到内存后的页面包含可读属性
31	IMAGE_ SCN_ MEM WRITE或80000000h	映射到内存后的页面包含可写属生